Part D — AI-Augmented Task

Prompt Used

"Explain Logistic Regression with Python example using sklearn on SUV dataset."

AI Output

The AI returned the following pipeline:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

df = pd.read_csv('suv_data.csv')
df['Gender'] = LabelEncoder().fit_transform(df['Gender'])

X = df[['Age', 'EstimatedSalary']]
y = df['Purchased']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

model = LogisticRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print('Accuracy:', accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
```

Evaluation

The code is correct and covers all the required steps. The scaler is fit only on the training set and applied separately to the test set, which is the right approach — fitting on both would leak test distribution into training.

One thing it skips is the optional decision boundary plot, but that was added separately in Part B. Everything else was tested and ran without errors on the dataset.